# Exam 1 Study Guide

**STAT 251 SECTION 01**

**2026-09-14**

# Exam Logistics:

- The exam is scheduled for **Friday, September 18, 2026**, and will be taken in class. An in-class review will be held on **Wednesday, September 16, 2026**
- You will have the entire 50 minute class period to complete the exam
- The exam is proctored in class but delivered through **Canvas**, so you must bring a fully charged laptop. **Notes are not permitted.** A calculator is permitted, but it must not be capable of wireless communication
- The exam will be about 20 - 25 questions long, with at least one bonus question. Questions are mostly multiple choice, short answer, or labeling problems, with the possibility of one or two short essay / longer explanation problems

# How to prepare

The exam asks you to **interpret** values, **identify** the correct equation, and **choose** the right summary for a distribution. Long calculations are not required: where a number is needed it is given, or the arithmetic is simple enough to do in your head.

Study by computing anyway. Work the Exam 1 Review Problems (../Review_problems/exam1_review_problems.html) through, then check the solutions (../Review_problems/exam1_review_problems_solns.html). They are more computational than the exam by design.

# Content To Study

- Know the “anatomy” of data (i.e., observations and variables, quantitative and categorical variables).
- Be able to identify the different types of quantitative and qualitative variables.
- Understand samples versus populations.

- Understand statistics versus parameters.
- Understand descriptive versus inferential statistics.
- What is meant by a distribution?
- Be able to construct a frequency table from a small set of observations and know how to find the frequency, relative frequency, and cumulative relative frequency.
- Know how to construct a dot plot, stem plot, and a histogram given a small set of observations. Where a histogram or a binned frequency table is involved, **the bins will always be given to you** – you will never be asked to choose the number of bins or the bin width yourself.
- Be able to compute a mean, median, and mode given a small set of observations. Note that you should also know how to compute a mean using a frequency table.
- Know how to compute a proportion.
- Know how to find the modal category of a qualitative variable.
- Be able to compute and interpret the variance and standard deviation given a small set of observations.
- Know how to compute the five-number summary of a variable.
- Quartiles on this exam use the **median of halves** method we use in class: find the median first, and when $n$ is odd leave that median out of both the lower and the upper half.
- Understand how to plot a cumulative distribution and use it to find the percentiles or quartiles of a distribution.
- Know how a box plot is constructed from a five number summary.
- Know how to interpret the shape (symmetry, skew, modality) of a distribution and how it is related to the mean and median.
- Know how to use the $1.5 \times IQR$ Rule to identify outliers.
- Understand what it means to say that a summary measure is resistant to outliers, and which summary measures we have discussed that are resistant and which are not.
- Understand the definition of a percentile and be able to interpret percentiles from a cumulative distribution plot or frequency table, and know that the $50^{th}$ percentile is the median and the $25^{th}$ and $75^{th}$ are $Q1$ and $Q3$.
- Be sure you understand the notation (i.e., symbols) we have used so far (e.g., $n, N, s, s^2, \bar{x}, \mu, \sigma, \sigma^2, \hat{p}, p$).

# Formulas on the exam

Notes are not permitted. The formulas below will be displayed on the projector during the exam, **unlabeled** – the equations only, with no names or descriptions.

You must be able to identify each one: the population mean from the sample mean, the population proportion from the sample proportion, the variance from the standard deviation. Several questions depend on it.

Look closely at the statistic/parameter pairs below. In each pair the *arithmetic is identical* and only two things change: the symbol on the left, and whether the count on the bottom is $n$ (a sample) or $N$ (a whole population).

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\bar{x} = \frac{1}{n}\sum_{x} xF(x), \quad \bar{x} = \sum_{x} xRF(x)$$

$$\hat{p} = \frac{\text{number of observations in the category}}{n}$$

$$p = \frac{\text{number of observations in the category}}{N}$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \quad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\text{range}(x) = \max(x) - \min(x), \quad IQR = Q3 - Q1$$

$$x < Q1 - 1.5 \times (Q3 - Q1), \quad x > Q3 + 1.5 \times (Q3 - Q1)$$

Self-test: cover the left-hand side of each line above and name what is on the right.